Shunting Inhibition and Dendritic Branching Shape Local Credit Assignment

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Abstract

Biological neurons assign credit through structures that standard neural networks usually ignore: synapses are distributed across branching dendrites, excitatory and inhibitory inputs evoke synaptic conductances, and inhibition can shunt local voltage by changing total input resistance. We study conductance-based dendritic networks with E/I synapse banks, shunting inhibition, and tree-structured branch-to-soma coupling, and ask when a low-bandwidth somatic broadcast can replace compartment-specific backpropagated errors. Exact gradients for these dendritic trees show that each synaptic update separates into a local eligibility term built from presynaptic activity, driving force, and input resistance, and one non-local compartment error. Under identity upward transfer, that error is the somatic error transported through conductance-stage path gains. This factorization turns local learning into a credit-signal compression problem: broadcast is useful only when it approximates the path-specific compartment-error field. We test this mechanism across gradient, oracle-transport, inhibition-intervention, competence, morphology, and harder-data diagnostics. Shunting improves LocalCA to backprop alignment, reduces gradient-scale mismatch, and makes exact compartment errors more rank-1 compressible than matched additive controls in the regimes studied here. Inhibitory-conductance interventions show that learned inhibition carries sample-specific pathway structure, while transported-error oracle experiments show that restoring the exact effective path-gain error largely closes the local-learning gap. Under nonnegative conductances and rank-1 broadcast, 5F shunting LocalCA remains within around 5 percentage points of matched backpropagation reference performance on MNIST, Fashion-MNIST, and context gating. These results suggest that E/I conductance, shunting inhibition, and dendritic branching can reshape credit-signal geometry and make low-bandwidth local learning more faithful.

1 Introduction

Unlike point units in standard neural networks, biological neurons assign credit across branching dendritic trees. A synapse does not only see presynaptic activity and a scalar postsynaptic activation; it also has access to local membrane voltage, synaptic reversal potential, total conductance, and input resistance. Inhibitory input can therefore do more than subtract activity: by increasing local conductance, it can shunt voltage and change the gain of every dendritic path passing through that compartment.

We examine whether these biophysical ingredients can facilitate local credit assignment (LocalCA). Backpropagation [10] would assign a distinct error to every dendritic compartment, whereas a biologically plausible supervisory signal is more likely to arrive as a low-bandwidth somatic or modulatory broadcast [15]. The central question is therefore not whether such a broadcast can reproduce unconstrained backpropagation in every setting, but when the exact dendritic error field is simple enough for it to approximate. Our contribution is not simply to add dendritic structure to a neural network, but to show how conductance, shunting inhibition, and dendritic topology change the geometry of the credit signal itself.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1). Each synaptic gradient factorizes into a synapse-local eligibility term and a single

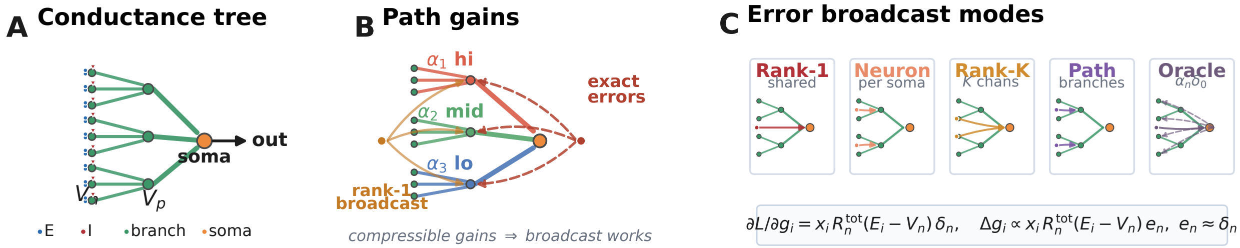

Figure 1: **Model and credit assignment. (A)** Conductance tree with nonnegative excitatory drive and inhibitory shunting load. **(B)** Exact credit varies by dendritic path, whereas rank-1 LocalCA broadcasts one shared error field. **(C)** LocalCA preserves the exact synapse-local eligibility and substitutes $e_n \approx \delta_n$; rank-1, neuron-wise, rank-K, and path-structured broadcasts trade biological bandwidth for fidelity, while transported error is an oracle.

non-local compartment error. The eligibility term depends on presynaptic activity, driving force, and input resistance. The compartment error is path-specific: under identity upward transfer it is the somatic error multiplied by a conductance-stage path gain through the dendritic tree. LocalCA preserves the exact local eligibility and replaces this path-specific error with a broadcast field.

This factorization turns local credit assignment into a compressibility problem. By compressibility, we mean that the matrix of exact compartment errors across examples and compartments is well approximated by a low-rank signal, especially a rank-1 field shared across compartments. Dendritic branching creates the paths; E/I synapse banks set the local conductance state; and shunting inhibition changes the input resistance along each path. Shunting is not assumed to help monotonically. It helps rank-1 broadcast when inhibitory conductance attenuates high-gain paths or otherwise makes the exact error field more compressible.

Related work. This work builds on models of dendritic integration, branch-level nonlinear computation, normalization, and biologically plausible teaching signals [29, 1, 3, 4, 30, 6, 5, 12, 13, 14, 21, 22, 23, 17, 40]. It also connects to machine-learning approaches that replace exact backpropagation with local, random, predictive, equilibrium, perturbation, or broadcast feedback signals [10, 9, 11, 26, 16, 27, 28, 24, 25, 31, 32, 33, 34, 35]. Our contribution is complementary: we derive the exact conductance-tree credit object and test when shunting inhibition and dendritic morphology make that object broadcast-compatible.

We test this chain directly. Exact-gradient reconstruction verifies the factorization numerically. Gradient-fidelity diagnostics show that shunting produces local updates that are better aligned and better scaled relative to backpropagation than matched additive controls. Exact-error compressibility and inhibition-intervention diagnostics show that learned inhibitory conductance carries sample-specific pathway structure. Transported-error oracle experiments show that supplying the exact effective path-gain-transported compartment error largely closes the local-learning gap, identifying broadcast fidelity as an important bottleneck. In an MNIST gradient-dynamics diagnostic, shunting improves final LocalCA to backprop alignment (cosine 0.100 ± 0.033 vs. 0.005 ± 0.026) and reduces scale mismatch (0.26 ± 0.05 vs. 0.50 ± 0.05). On noise resilience under matched inhibition (at least five inhibitory synapses per branch), transported error raises additive local learning from 81–84% under rank-1 broadcast to 92–94%, nearly matching shunting at 95–95.2%.

Contributions. This paper makes three contributions: deriving the exact conductance-tree gradient and its local-eligibility/path-error factorization; identifying path-gain compressibility as the quantity mediating whether low-rank somatic broadcast can approximate that error; and showing empirically that shunting inhibition can improve this approximation while also identifying regimes where rank-1 feedback fails and richer pathway-specific feedback is needed.

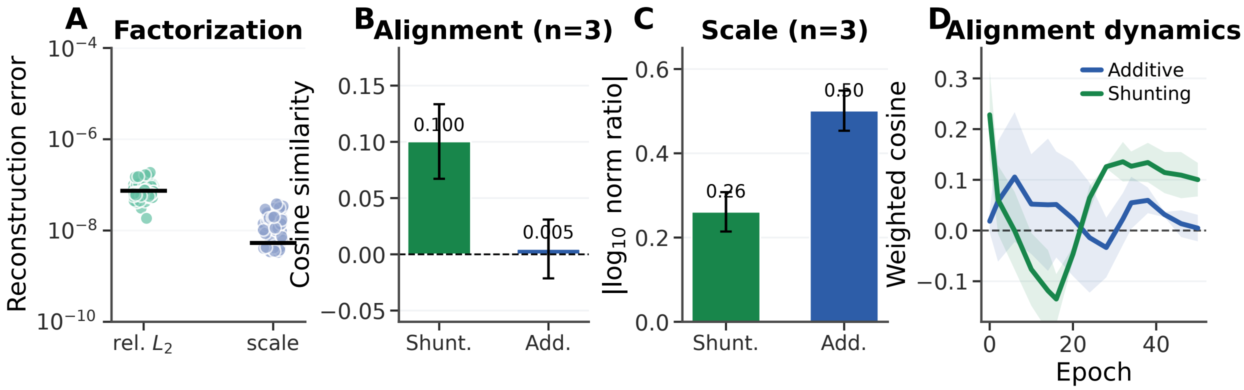

Figure 2: **Gradient-fidelity diagnostic.** **(A)** Exact-factorization sanity check: reconstructed gradients match autograd to numerical precision. **(B)** Final-state weighted cosine similarity between local and backprop gradients. **(C)** Final-state scale mismatch, $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$. Error bars are ± 1 s.d. across 3 gradient-dynamics seeds; main learning and oracle sweeps use 5 seeds unless noted. **(D)** Alignment over learning; Appendix Fig. S2 shows finite gradient norms.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal potentials are set to 0, the excitatory reversal potential is set to 1, and leak conductance is fixed at 1 (i.e., all conductances are expressed relative to leak). Each compartment computes a conductance-weighted voltage: excitatory conductance pulls the voltage toward the excitatory reversal potential, leak and shunting inhibition pull it toward zero, and dendritic coupling transfers child voltages upward toward the soma. This form makes local sensitivity depend on both driving force $(E-V)$ and total input resistance R^{tot}. Throughout the paper we write dendritic morphologies as rooted trees $[b_1, b_2, \ldots, b_D]$, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus $[3,3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward experiments, synaptic inputs terminate on dendritic branches rather than directly on the soma. Each branch layer carries explicit excitatory conductance banks and, when enabled, learned inhibitory branch-level conductance banks driven by nonnegative inputs. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit; explicit inhibitory-cell controls are reported in the appendix. The additive and shunting cores are architecture-matched at the tree level and differ only in the branch-level voltage integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in path gain through input resistance. The local learning rule then asks whether a low-bandwidth teaching signal can stand in for the exact path-specific error induced by that tree. **Notation.** We use α_n^{cond} for the conductance-stage path gain from compartment n to the soma and $\tilde{\alpha}_n$ for the effective path gain after including any post-voltage activation derivatives; when upward transfer is identity, $\tilde{\alpha}_n = \alpha_n^{\text{cond}}$. We use N_E and N_I for the number of excitatory and inhibitory synapses per branch, $r_n^{4\text{F}}$ for the empirical 4F branch-soma covariance proxy, ϕ_n for the 5F confidence factor, λ_{mix} for the pathway-broadcast mixing weight, and κ_j for depth-dependent modulation in morphology extensions. All conductances and presynaptic activities in the main experimental setting are nonnegative; additive voltages may be signed comparators, but the conductance variables that define the shunting model are nonnegative.

Voltage equation and local sensitivities. Consider compartment n with synaptic inputs j (activity x_j, reversal E_j, conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j, conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \qquad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \tag{1}$$

V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathscr{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathscr{S}_n \leq V_n \leq \max \mathscr{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow directly:

Proposition 1 (Local Sensitivities)**.**

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}}(E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}}(V_i - V_n). \tag{2}$$

Eq. (2) gives the eligibility factors used by the local rules below: presynaptic activity or voltage difference, driving force, and input resistance.

Shunting inhibition as divisive gain control. An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n) x_j g_j^{\text{syn}}$ and increases g_n^{tot}. Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned inhibitory conductances serve an analogous role. The same denominator also gives the direct connection to path gain. Writing $G_n^E(x)$ and $G_n^I(x)$ for the total excitatory and inhibitory synaptic conductance impinging on compartment n,

$$R_n^{\text{tot}}(x) = \left(1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}}\right)^{-1}.$$

Increasing $G_n^I(x)$ lowers $R_n^{\text{tot}}(x)$ and therefore multiplicatively suppresses every upstream path whose error must pass through compartment n. In this sense inhibition can gate credit flow by changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2 formalizes this statement after the tree path gain is defined.

Exact gradients for dendritic trees. Let V_0 be the somatic output and define the soma/core error as $\delta_0 := \partial L / \partial V_0$. For a linear decoder $\hat{y} = W_{\text{dec}} V_0$, this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$; for a nonlinear decoder, δ_0 is the corresponding decoder-input Jacobian product. The theoretical derivation is stated for the conductance-stage voltage before any post-voltage reactivation. When a branch applies a monotone reactivation before transmitting activity upward, the same recursion holds with additional local derivative factors along the path. All exact-error diagnostics below are computed as pre-reactivation voltage errors, including activation-derivative factors when reactivation is enabled.

Theorem 1 (Conductance-Stage Backpropagation on a Dendritic Tree)**.** *For a rooted dendritic tree with soma at node* 0 *and unique parent $p(n)$ for each non-somatic compartment, suppose the parent receives either the pre-reactivation voltage V_n or a local post-voltage activity $a_n = f_n(V_n)$. For identity upward transfer, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} \, g_{n \to p(n)}^{\text{den}}, \tag{3}$$

with boundary condition $\partial L / \partial V_0 = \delta_0$. With post-voltage transfer $a_n = f_n(V_n)$, the local factor becomes $f_n'(V_n) R_{p(n)}^{\text{tot}} g_{n \to p(n)}^{\text{den}}$. Because the graph is a tree, each compartment has a unique path to the soma. Defining the conductance-stage path gain

$$\alpha_n^{\text{cond}} = \prod_{(i \to k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} \, g_{i \to k}^{\text{den}}, \qquad \frac{\partial L}{\partial V_n} = \alpha_n^{\text{cond}} \delta_0, \tag{4}$$

with $\alpha_0^{\mathrm{cond}} = 1$ under identity transfer. In the post-voltage case, the exact effective path gain is

$$\tilde{\alpha}_n = \prod_{(i\to k)\in \mathrm{path}(n\rightsquigarrow 0)} f_i'(V_i)\, R_k^{\mathrm{tot}}\, g_{i\to k}^{\mathrm{den}}, \qquad \frac{\partial L}{\partial V_n} = \tilde{\alpha}_n \delta_0, \tag{5}$$

where $f_i'(V_i)$ is omitted for identity transfer.

Proof. Apply the chain rule on the tree-structured computation graph using Prop. 1. □

The one-step recursion in Eq. (3) and the path product in Eq. (4) separate local conductance transfer from the non-local somatic error.

Proposition 2 (Inhibitory control of conductance-stage path gain)**.** *Let $\mathscr{A}(n)$ be the set of compartments on the path from compartment n to the soma, excluding n and including the parent compartments through which the error must pass. Holding dendritic coupling conductances fixed, an added inhibitory conductance $\Delta G_k^I(x) \geq 0$ at any $k \in \mathscr{A}(n)$ changes the conductance-stage path gain by*

$$\frac{\alpha_n^{\mathrm{cond}}(x;\Delta G^I)}{\alpha_n^{\mathrm{cond}}(x;0)} = \prod_{k\in\mathscr{A}(n)} \frac{g_k^{\mathrm{tot}}(x)}{g_k^{\mathrm{tot}}(x)+\Delta G_k^I(x)}, \qquad \frac{\partial \log \alpha_n^{\mathrm{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\mathrm{tot}}, & k\in\mathscr{A}(n), \\ 0, & k\notin\mathscr{A}(n). \end{cases} \tag{6}$$

Proof. Substitute $R_k^{\mathrm{tot}} = 1/g_k^{\mathrm{tot}}$ into the path-gain product in Eq. (4). Adding inhibitory conductance changes only the denominator at compartments on the path from n to the soma, which gives the product ratio and the log derivative in Eq. (6). □

Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths below the inhibited compartment. Second, inhibition is useful for rank-1 local learning only when this attenuation makes the distribution of α_n^{cond} more uniform or better aligned with the task. If inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population, it can suppress useful signals and hurt learning. In the additive control the same I input changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as a multiplicative resistance gate.

When does shunting compress path gains? The attenuation identity above is not by itself a concentration result. Let $z_n = \log \alpha_n^{\mathrm{cond}}$ be the log conductance-stage path gain for compartment n. If added inhibitory conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k\in\mathscr{A}(n)} \log\left(1+\frac{\Delta G_k^I(x)}{g_k^{\mathrm{tot}}(x)}\right), \qquad z_n' = z_n - \beta_n, \tag{7}$$

then, over compartments or examples,

$$\mathrm{Var}(z') = \mathrm{Var}(z) + \mathrm{Var}(\beta) - 2\,\mathrm{Cov}(z,\beta). \tag{8}$$

Thus shunting narrows the log path-gain field exactly when

$$\mathrm{Cov}(z,\beta) > \frac{1}{2}\mathrm{Var}(\beta). \tag{9}$$

Eq. (7) follows by taking the logarithm of the product ratio in Prop. 2; Eq. (8) then applies the variance identity for $z-\beta$. Eq. (9) is an interpretive sufficient condition under the same averaging measure, not a primary empirical diagnostic. In finite trained networks, the covariance margin can disagree with CV and exact-error-rank summaries because these statistics average over different compartment and sample axes. We therefore use direct path-gain dispersion, exact-error compressibility, and inhibition-intervention measurements as the empirical tests, and report the covariance margin as a boundary check in the appendix.

Corollary 1 (Local–Global Factorization)**.** *The exact synaptic gradient at compartment n factorizes as:*

$$\frac{\partial L}{\partial g_i^{\mathrm{syn}}} = \underbrace{x_i R_n^{\mathrm{tot}}(E_i - V_n)}_{\textit{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\textit{compartment error}}, \tag{10}$$

where the eligibility term depends only on quantities available at synapse i (presynaptic activity x_i, input resistance R_n^{tot}, driving force $E_i - V_n$), and the compartment error $\partial L/\partial V_n$ is the sole non-local quantity.

The factorization in Eq. (10) isolates the only non-local quantity as $\partial L/\partial V_n = \alpha_n^{\mathrm{cond}}\delta_0$ under identity upward transfer and $\partial L/\partial V_n = \tilde{\alpha}_n\delta_0$ when post-voltage transfer is enabled. The compartment error remains the only non-local term. Any practical local rule therefore depends entirely on how well a broadcast signal approximates that exact compartment error, which we evaluate in Sec. 4. In the models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot}, which narrows the spread of local sensitivities across the tree. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation $f(V_n)$ before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation.

3 Local Learning Rules

Broadcast error approximation. We approximate the exact compartment error $\partial L/\partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic/core error δ_0. For a linear readout this reduces to $\delta_0 = W_{\mathrm{dec}}^{\top}(\partial L/\partial\hat{y})$; for a nonlinear decoder we use the corresponding decoder-input Jacobian product $J_{\mathrm{dec}}(h_{\mathrm{core}})^{\top}(\partial L/\partial\hat{y})$ so that the broadcast remains defined in the same core-space coordinates. For rank-1 broadcast, $\bar{\delta}_b = d_0^{-1}\sum_{c=1}^{d_0}\delta_{0,bc}$ is computed separately for each example b and broadcast to all compartments of that example. We distinguish five feedback objects: **rank-1 shared broadcast**, $e_n = \bar{\delta}\mathbf{1}$; **neuron-wise broadcast**, $e_n = \delta_0$ when layer width matches the error dimension; **rank-K random broadcast**, $c = P_K\delta_0$ and $e_n = \sum_k q_{n,k}c_k$; **path-structured broadcast**, $e_n = \lambda_{\mathrm{mix}}\bar{\delta}\mathbf{1} + (1-\lambda_{\mathrm{mix}})\Gamma_n(\delta_0)$ with pathway roles motivated by vectorized dendritic teaching signals [39]; and the **transported oracle**, $e_n^{\mathrm{transport}} = \tilde{\alpha}_n\delta_0 = \partial L/\partial V_n$, an analysis upper bound because it supplies the effective path gain from Theorem 1. Operationally, path-transport runs with reactivation enabled transport activation-space errors through parent reactivation slopes and multiply by the local $f_n'(V_n)$ before conductance updates; with identity transfer this reduces to $\alpha_n^{\mathrm{cond}}\delta_0$. In the main feedforward architectures, hidden widths exceed the decoder-error dimension, so the default neuron-wise broadcast reduces to a rank-1 shared field (Appendix Table S9). Thus the main rank-1 result should be read as the architecture-induced low-bandwidth case, not as a claim that neuron-wise broadcasts were unavailable in principle. Under shunting, that deliberately low-bandwidth field is often sufficient on standard classification tasks. Routed tasks and rank-bridge controls in the appendix test the opposite regime, where rank-1 broadcast collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

Three-factor learning rule (3F).

Definition 1 (3F Update)**.** *For synaptic and dendritic conductances:*

$$\Delta g_j^{\mathrm{syn}} = \eta\langle x_j R_n^{\mathrm{tot}}(E_j - V_n)\,e_n\rangle_B, \qquad \Delta g_j^{\mathrm{den}} = \eta\langle R_n^{\mathrm{tot}}(V_j - V_n)\,e_n\rangle_B, \tag{11}$$

where $\langle\cdot\rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n. The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$. We write Δg as the local gradient estimate supplied to the optimizer; the optimizer applies the usual descent step. Equivalently, one could absorb the minus sign into e_n and treat it as a descent signal.

Additive control. Rule (11) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive E/I control, inhibition enters as a fixed signed voltage contribution rather than through the conductance denominator:

$$V_n^{\text{add}} = \sum_{j \in E} g_j^E x_j^E - \sum_{j \in I} g_j^I x_j^I + \sum_k g_k^{\text{den}} V_k.$$

Equivalently, $\partial V_n^{\text{add}} / \partial g_j^{\text{syn}} = s_j x_j$ with $s_j = +1$ for excitatory and $s_j = -1$ for inhibitory synapses, and $\partial V_n^{\text{add}} / \partial g_k^{\text{den}} = V_k$. The additive local gradient therefore has no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model. In implementations with unconstrained raw parameters θ and $g = \text{softplus}(\theta)$, Eqs. (11)–(12) give conductance-space gradients; before assigning gradients to θ, we multiply by $\partial g / \partial \theta$. The exact-gradient reconstruction diagnostics include this parameterization factor.

Practical heuristic wrappers: 4F and 5F. 4F (branch-soma relevance proxy). In the conductance-stage recursion (4), the path gain α_n^{cond} attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$r_n^{\text{4F}} = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n)\,\text{Var}(\bar{V}_0)} + \varepsilon).$$

High r_n^{4F} up-weights branches whose voltages covary with the soma; low r_n^{4F} down-weights branches where the broadcast is likely to be a poor proxy. We keep 4F as a diagnostic intermediate. Empirically, this factor alone gives little improvement over 3F; the practical jump comes from the 5F confidence factor below.

5F (confidence-weighted 4F). 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally, ϕ_n is a branch-wise confidence gate, clamped to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived; its gain should be read as practical denoising rather than theoretical necessity.

Definition 2 (5F Update)**.**

$$\Delta g_j^{\text{syn}} = \eta\, r_n^{\text{4F}} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n)\, e_n \rangle_B, \qquad \Delta g_j^{\text{den}} = \eta\, r_n^{\text{4F}} \phi_n \langle R_n^{\text{tot}} (V_j - V_n)\, e_n \rangle_B. \tag{12}$$

Eq. (12) is therefore the main practical LocalCA update, while the 3F rule is the theoretically derived local gradient and 4F is the intermediate ablation. A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

4 Experiments

Setup. The experiments test five linked predictions: exact factorization should reconstruct autograd gradients; shunting should improve LocalCA direction and scale relative to backpropagation; exact compartment errors should be more compressible under shunting than under matched additive controls; learned inhibition should carry sample-specific pathway structure rather than merely adding conductance load; and replacing the broadcast with effective transported error should largely close the local-learning gap. We evaluate capacity-calibrated supervised tasks and controlled sweeps that isolate inhibition strength, rule family, broadcast mode,

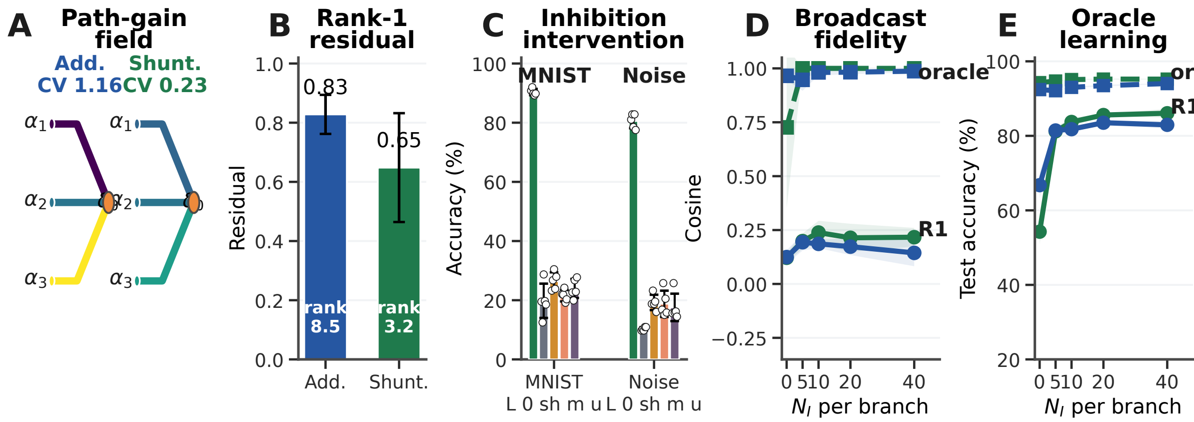

Figure 3: **Mechanistic chain from path gains to learning.** **(A)** Dendritic path-gain fields at matched inhibition. **(B)** Exact-error compressibility on MNIST: lower rank-1 residual and participation rank favor scalar broadcast. **(C)** Post-training inhibitory-conductance interventions (L: learned; 0/sh/m/u: zero, shuffled, mean-clamped, uniform matched). **(D)** Compartment-error fidelity on noise resilience; dashed lines show the transported-error oracle. **(E)** Transported-error learning recovers near-ceiling local learning where rank-1 broadcast fails. Error bars denote ± 1 s.d. across seeds.

morphology, and architecture. The main text uses MNIST [37], Fashion-MNIST [36], context gating, noise resilience, and a compressed morphology-regime summary; CIFAR-10 [38] and routed-feedback diagnostics are reported in the appendix (Figs. S6, S10). Main comparisons are architecture-matched additive vs. shunting dendritic cores and local vs. backprop training; DFA is a supplementary baseline. Unless otherwise stated, local runs use 5F with the default neuron-wise broadcast, which reduces to rank-1 shared broadcast when hidden width exceeds the decoder-error dimension. All main-claim dendritic runs use nonnegative conductance transforms and nonnegative first-layer synaptic drive; image inputs are in $[0,1]$, corrupted inputs are clipped back to $[0,1]$, and context gating uses a nonnegative contextual figure-ground MNIST construction. The transported-error condition is an oracle upper bound derived from Theorem 1, not the proposed biological rule. Main mechanistic, oracle, and competence results use 5 seeds per condition unless noted otherwise. For gradient diagnostics, weighted cosine is $\sum_p w_p \cos(g_p^{\text{local}}, g_p^{\text{bp}})$ with $w_p = \text{numel}(p)/\sum_q \text{numel}(q)$, and weighted scale mismatch uses the same weights on $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$. Path-gain CV is the coefficient of variation of α_n^{cond} over the measured compartment field. Exact-error compressibility uses a matrix E whose rows are examples and whose columns are compartment–neuron coordinates; if σ_i are its singular values, the rank-1 residual is $\rho_1 = \sqrt{\sum_{i>1}\sigma_i^2}/\sqrt{\sum_i \sigma_i^2}$ and participation rank is $r_{\text{part}} = (\sum_i \sigma_i^2)^2/\sum_i \sigma_i^4$.

From exact factorization to credit-signal fidelity.

To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p, we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

Across the MNIST gradient-dynamics diagnostic, shunting is more aligned with backprop and closer in norm than the additive control at the final checkpoint: weighted cosine rises from 0.005 ± 0.026 to 0.100 ± 0.033, and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05. Figure 2A confirms that the implementation matches the theory under the nonnegative-input setup: reconstructing gradients from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below 2×10^{-7} and weighted scale mismatch below 4×10^{-8} across runs. In a few low-norm parameter blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact, so we report the norm-based diagnostics

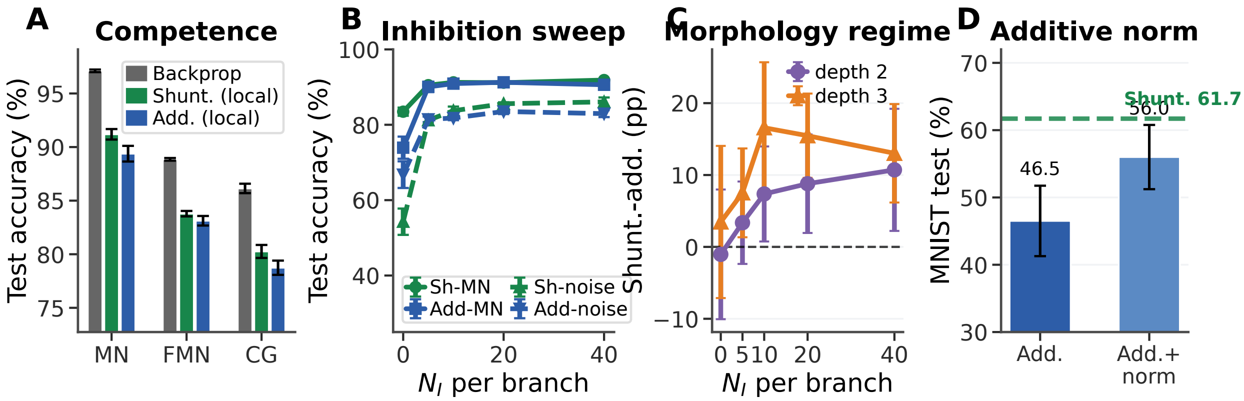

Figure 4: **Capacity-calibrated competence and regime dependence.** **(A)** Matched backprop reference performance vs. rank-1 5F LocalCA on MNIST, Fashion-MNIST, and context gating. **(B)** Inhibitory dose-response. **(C)** Morphology-dependent shunting advantage on noise resilience from a 3-seed sweep. **(D)** Separate stress-regime additive-normalization diagnostic: normalization improves additive LocalCA but remains below the shunting reference. Error bars denote ± 1 s.d. where available.

in the figure. Figure 2D shows the same gradient-alignment comparison over training: additive alignment remains near zero at the final checkpoint rather than failing only at one static snapshot. Appendix Fig. S2 verifies that the corresponding local and backprop gradient norms remain finite throughout learning, ruling out a trivial zero-gradient explanation. At the level of the theoretically derived 3F rule, the central non-local approximation is therefore the broadcast substitute for $\partial L/\partial V_n$.

Mechanistic chain: path gains, broadcast fidelity, and oracle transport. Figure 3 provides the central empirical chain. At matched inhibition (N_I=5), shunting narrows the MNIST conductance-stage path-gain field (CV 0.23 vs. 1.16 for additive) and makes the exact compartment-error matrix more compressible: the best rank-1 residual drops from 0.83 ± 0.07 to 0.65 ± 0.18, while participation rank drops from 8.5 ± 3.3 to 3.2 ± 1.2. This is the diagnostic matched to the paper's main claim: rank-1 broadcast is useful when the exact error field is closer to low rank.

The inhibition intervention shows that the learned inhibitory field is necessary for the trained computation and is not interchangeable with a matched conductance load: zeroing, shuffling, or clamping learned branch inhibition drops MNIST from about 90% to 20–26% and noise resilience from about 81% to 10–19%. The fidelity and oracle panels probe the same broadcast bottleneck: rank-1 broadcast tracks the exact compartment error better under shunting once inhibition is present, transported error reaches about 0.95–1.0 fidelity, and transported learning lifts both cores to 92–95% on noise resilience. CIFAR-10 and direct-I versus explicit-I path-gain probes remain appendix diagnostics (Fig. S6; Table S4).

Capacity-calibrated competence. In the capacity-calibrated regime, shunting LocalCA remains within around 5 percentage points of matched backpropagation on MNIST, Fashion-MNIST, and context gating (Table S2; Fig. 4). In a separate lower-capacity stress-regime diagnostic, Fig. 4D shows that additive normalization improves additive LocalCA but remains below the shunting reference, so the gain is not explained by generic voltage normalization alone. Figure 5 summarizes the rule and feedback controls: 5F is the strongest practical optimizer, somatic rank-1 error is essential, and higher feedback bandwidth improves the noise-resilience diagnostic.

Regime dependence of the shunting advantage. With matched rank-1 broadcast, shunting is modestly better on MNIST, Fashion-MNIST, and context gating, including 0.803 ± 0.006 vs. 0.787 ± 0.007 for additive on context gating. The gap grows on inhibition-sensitive noise resilience once inhibitory conductance is present (Fig. 4B). Morphology shifts the useful inhibitory operating regime: different tree depths and branch

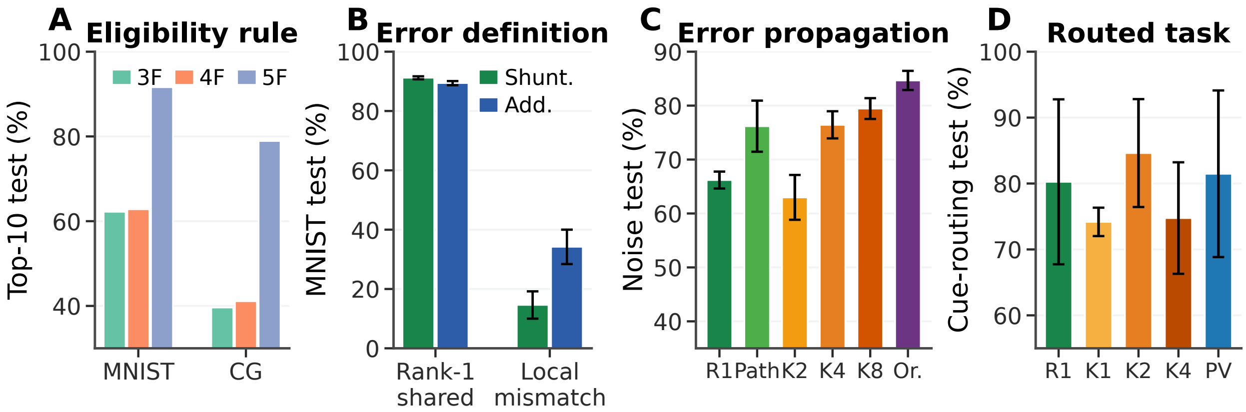

Figure 5: **Rule and feedback controls.** **(A)** Eligibility-rule ablation: 3F is the theorem-derived local eligibility, 4F adds branch–soma relevance, and 5F adds the confidence wrapper. **(B)** Error-source ablation: rank-1 shared somatic error works, whereas a local-mismatch surrogate collapses learning. **(C)** Feedback-bandwidth ladder on noise resilience: R1 is rank-1, Path is recursive path-propagation, K2/K4/K8 are random low-rank, and Or. is exact effective transported error. **(D)** Cue routing tests branch-specific credit: R1 is rank-1, K1/K2/K4 are random low-rank, and PV is pathway-vector feedback; higher rank helps but PV does not yet beat the best unstructured low-rank control.

factors peak at different N_I, and the shunting advantage is not monotonic in inhibition (Fig. 4C). Appendix Fig. S4 extends this boundary check to depth scaling and noisy error broadcasts.

5 Discussion and Limitations

We studied local credit assignment in dendritic networks with E/I conductance synapses, shunting inhibition, and tree-structured branching. Exact gradients separate synapse-local eligibility–presynaptic activity, driving force, and input resistance–from one path-specific compartment error. This makes LocalCA an error-approximation problem: the local factors are available at the synapse, but the teaching signal must approximate the exact dendritic error field.

The experiments support a coherent mechanism. Branching creates credit paths, E/I banks set each compartment's conductance state, and shunting modulates input resistance along those paths. In the regimes studied here, this makes exact errors more rank-1 compressible, improves LocalCA to backprop alignment, and reduces scale mismatch. Transported-error oracles largely close the local-learning gap, while inhibition interventions show that learned inhibitory conductance carries sample-specific pathway structure rather than only global conductance scale.

The limits are important. Shunting is not universal, and more inhibition is not always better. The practical 5F rule is an empirical stabilizer rather than a theorem-derived necessity; Eq. (9) is interpretive; activation shape, morphology, task structure, and broadcast rank all change the operating regime. We do not claim competitive CIFAR-10 performance, a recurrent inhibitory circuit model, or that scalar broadcast suffices for routed/pathway-specific tasks. The main model also uses input-driven shunting inhibition rather than recurrent lateral inhibition, so recurrent inhibitory circuits remain a future direction. The broader message is that biophysical dendrites can change not only representations, but also the geometry of credit assignment.

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A Supplementary Results

Appendix overview. The appendix is organized by how directly each result supports the main mechanism. It first reports rule, feedback, gradient, competence, and baseline controls; then gives harder-data, morphology, inhibition, and routed-feedback diagnostics; and finally provides implementation, theory, and task details needed for reproducibility.

Boundary diagnostics. Several controls are intentionally negative or mixed. The log-gain covariance margin from Eq. (9) is not used as a positive main result, and the pathway-vector feedback variant does not beat the best unstructured low-rank cue-routing control. These diagnostics delimit the claim rather than smoothing over failure modes.

Rule-Family Ranking

This table keeps the 3F/4F/5F optimizer distinction explicit: the theorem-derived eligibility is necessary for the story, but the reported LocalCA competence results use the empirical 5F stabilizer.

Dataset	Rule	Top-10 sweep valid	Top-10 sweep test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	**0.912**	**0.916**
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	**0.807**	**0.789**

Table S1: **Rule-family ranking.** Mean over the ten best validation-selected configurations from local-competence sweeps; this is not top-10 classification accuracy.

Local Competence and Regime Dependence

Extended Gradient Analysis and N_I Sweep Detail

Alignment dynamics are not a zero-gradient artifact. To check whether low additive alignment could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout training (weighted norm range 9.5×10^{-4}–9.5×10^{-3}), and the corresponding backprop gradients are also nonzero (2.3×10^{-5}–5.7×10^{-3}). The low additive cosine therefore reflects directional and scale mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018 to 0.005), while its local/backprop norm ratio falls from about 8.8×10^{2} to 0.69.

Verification and Reproducibility

Additional Stress Tests

FA/DFA Baseline Comparison

CIFAR-10 Results

Table S3 provides the exact values for the appendix CIFAR-10 figure. In this stronger compact direct-I-stream family, shunting performs well under standard training (49.5% vs. 48.3% for additive), so the harder-data

Dataset	BP ceiling	Shunting local	Additive local	BP gap
MNIST	0.971	0.912 ± 0.005	0.894 ± 0.007	5.9%
Fashion-MNIST	0.889	0.838 ± 0.003	0.831 ± 0.004	5.1%
Context gating	0.861	0.803 ± 0.006	0.787 ± 0.007	5.8%

Table S2: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with rank-1 broadcast and local decoder updates. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix C). Error bars on local values: ± 1 s.d. across 5 seeds.

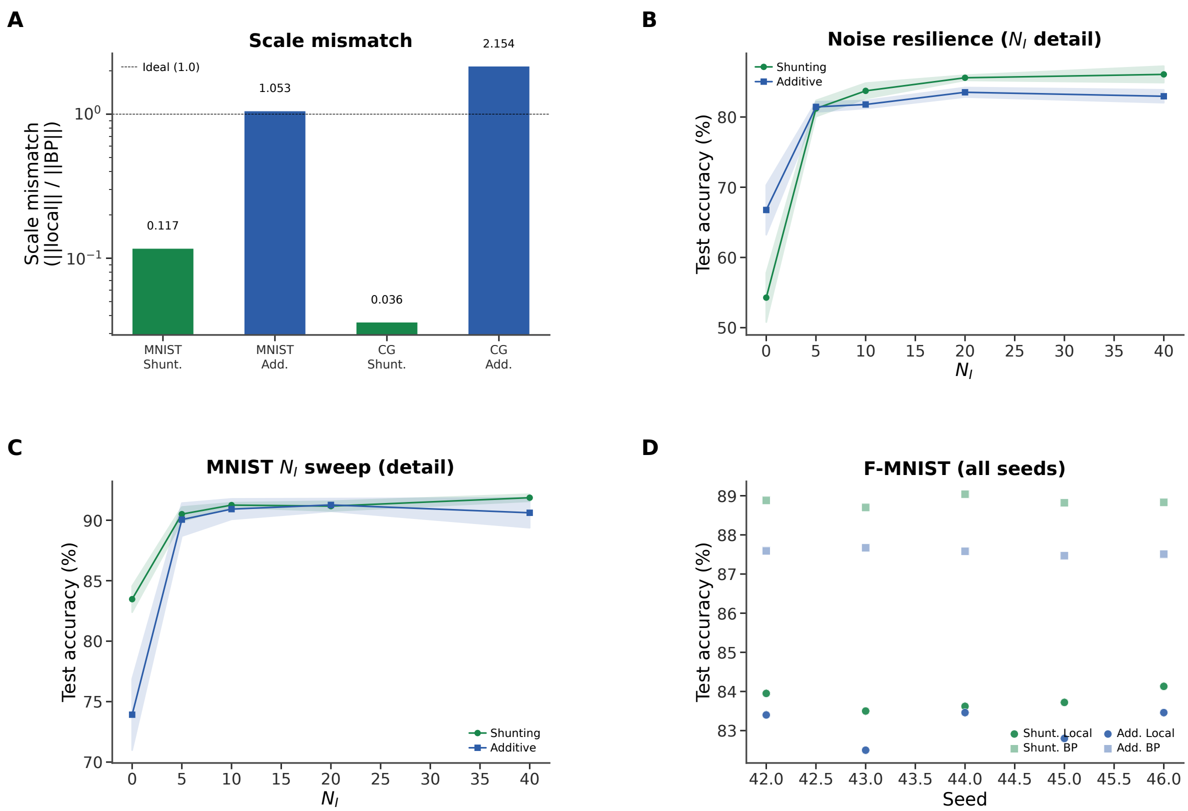

Figure S1: **Extended gradient and N_I analysis (supplementary). (A)** Scale mismatch bars from a separate matched-weight static diagnostic; these values should not be compared directly to the checkpointed final-state diagnostic in Fig. 2. Shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). **(B)** Noise-resilience N_I dose-response with error bands (± 1 s.d.). **(C)** MNIST N_I dose-response detail with error bands. **(D)** Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

stress test does not unfairly penalize the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware soma mapping is used: additive LocalCA reaches 25.5% under rank-1 per-soma broadcast and 32.4% under transported error, while shunting rises from 29.7% under rank-1 broadcast to 35.0% with random low-rank $K{=}4$ and 47.2% with transported error, nearly matching the 49.5% shunting backprop ceiling. Removing learned I-to-E conductance lowers the shunting ceiling by 5.7 pp and transported LocalCA by 8.6 pp, making this a harder-data diagnostic for inhibitory conductance as part of the useful credit geometry

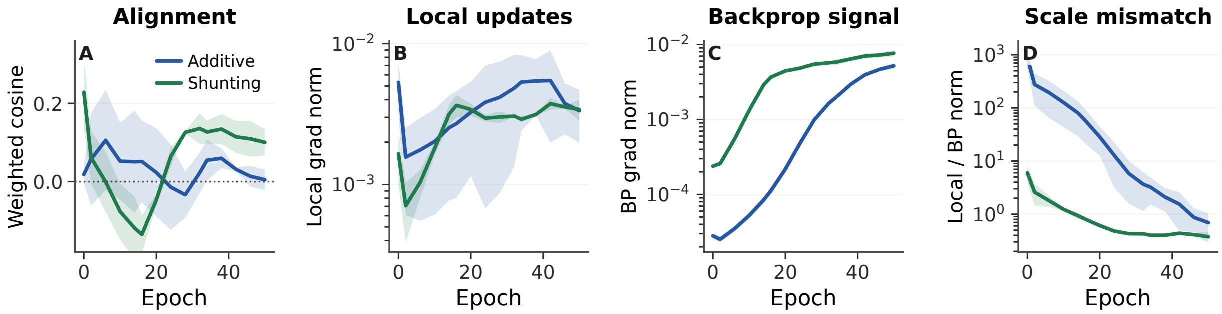

Figure S2: **Alignment dynamics with explicit gradient-norm checks.** **(A)** Weighted LocalCA to backprop cosine over training for the 5F rule. **(B,C)** Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. **(D)** The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

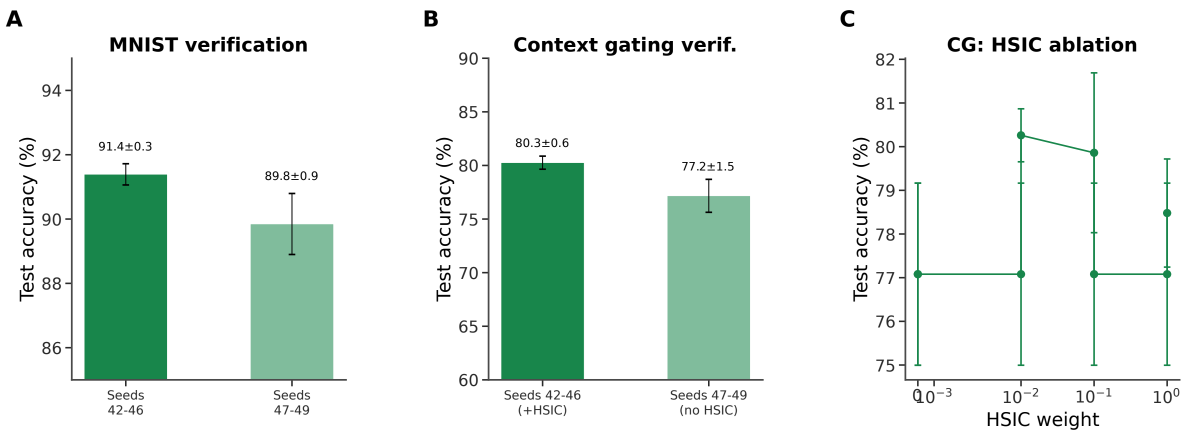

Figure S3: **Verification and reproducibility (supplementary).** **(A)** MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. **(B)** Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. **(C)** HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

rather than just an added architectural detail.

Additive Normalization Control

This control separates generic additive voltage normalization from shunting conductance dynamics in a lower-capacity MNIST stress regime.

Morphology $\times$ Inhibition Regime Map

This regime map tests whether tree geometry changes the inhibitory operating point at which shunting helps. It supports the claim that morphology changes the useful inhibition regime; a fuller mechanistic account would require matched theory diagnostics for every cell in the grid.

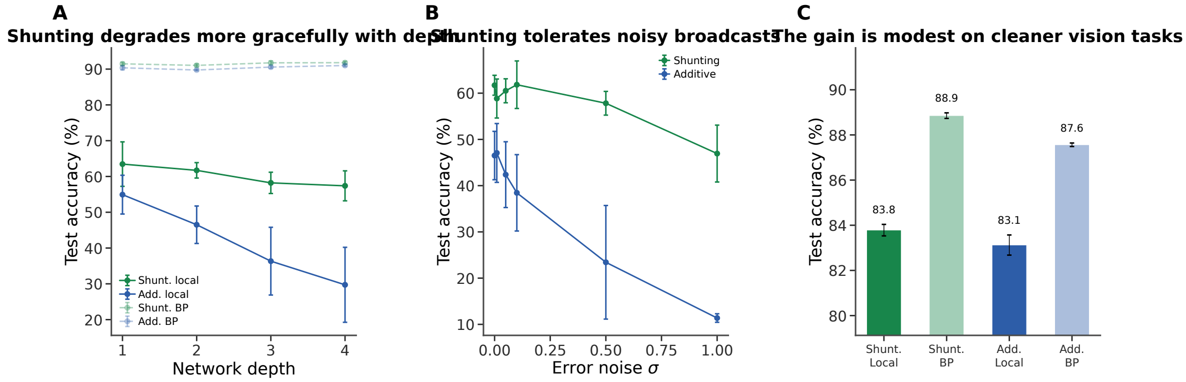

Figure S4: **Additional stress tests. (A)** Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. **(B)** Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. **(C)** Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

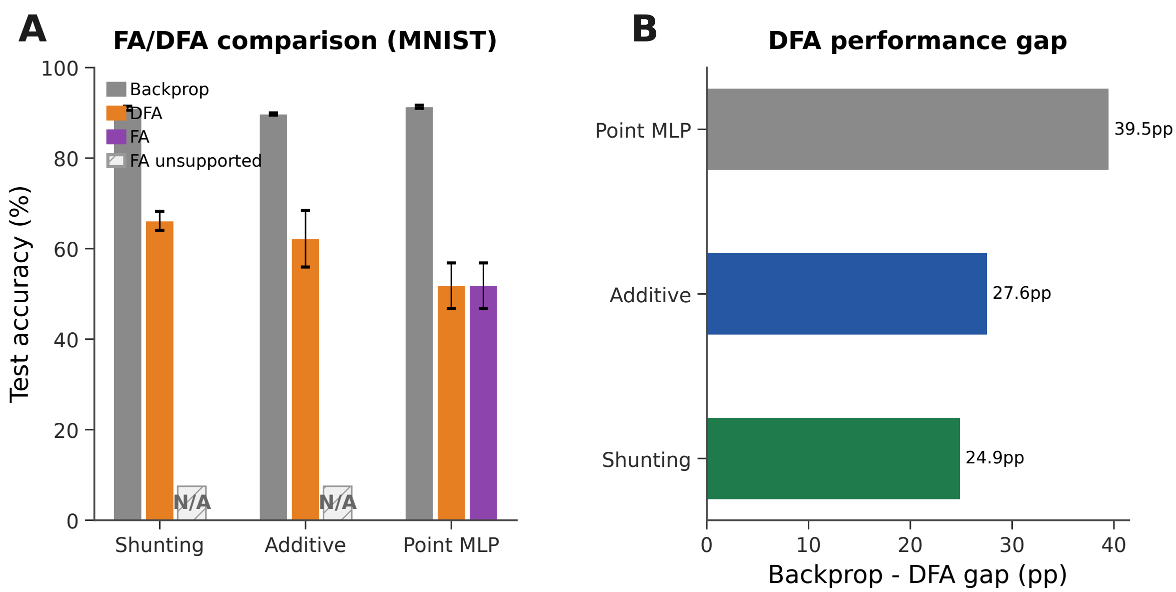

Figure S5: **Feedback alignment baselines (MNIST). (A)** Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. **(B)** Backprop−DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

The main experiments use input-driven inhibitory streams: the transfer layer provides nonnegative excitatory and inhibitory channels from the external input, and excitatory dendrites receive learned I-to-E conductance banks directly. To test whether the same mechanism can be generated by an explicit feedforward inhibitory

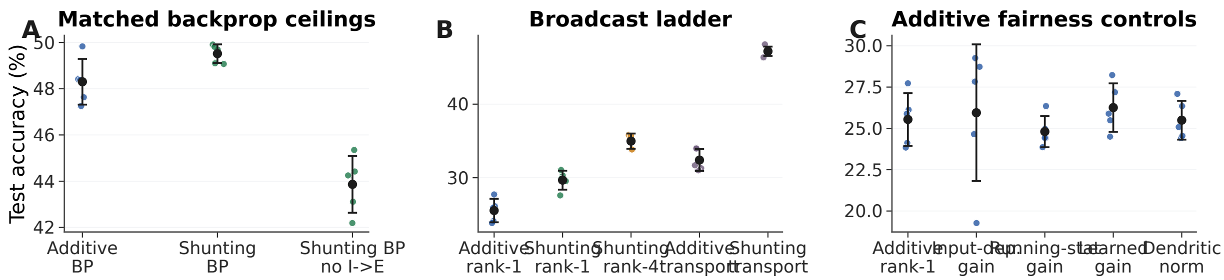

Figure S6: **CIFAR-10 control ladder in the compact direct-I-stream family.** **(A)** Matched backprop ceilings. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting ceiling. **(B)** Broadcast ladder. Rank-1 broadcast remains weak on this harder dataset, rank-4 improves partially, and exact effective transported error nearly reaches the shunting backprop ceiling. **(C)** Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive rank-1 broadcast. This figure is a harder-data diagnostic, not a claim of competitive CIFAR-10 benchmarking.

Condition	Core	Training / broadcast	I-to-E	Test acc.
standard	additive	backprop	yes	48.3 ± 1.0
standard	shunting	backprop	yes	$\mathbf{49.5 \pm 0.4}$
standard, no learned inhibition	shunting	backprop	no	43.9 ± 1.2
rank-1	additive	5F per-soma	yes	25.5 ± 1.6
rank-1, input-dependent gain	additive	5F per-soma	yes	25.9 ± 4.1
rank-1, running-stat gain	additive	5F per-soma	yes	24.8 ± 0.9
rank-1, learned gain	additive	5F per-soma	yes	26.3 ± 1.5
rank-1, dendritic normalization	additive	5F per-soma	yes	25.5 ± 1.2
rank-1	shunting	5F per-soma	yes	29.7 ± 1.3
rank-1, no learned inhibition	shunting	5F per-soma	no	22.3 ± 0.7
transported error	additive	5F path transport	yes	32.4 ± 1.5
rank-4	shunting	5F low-rank	yes	35.0 ± 1.0
transported error	shunting	5F path transport	yes	$\mathbf{47.2 \pm 0.6}$
transported error, no learned inhibition	shunting	5F path transport	no	38.6 ± 1.0

Table S3: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).** All rows use flattened CIFAR-10 in $[0,1]$, nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backprop ceiling, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive rank-1 fairness controls do not explain the transported-shunting result.

population, we ran a one-feedforward-layer noise-resilience probe that preserves the relevant dendritic depth ($[3,3]$ morphology). In this setting, explicit inhibitory cells reach $94.2 \pm 0.2\%$ under path-transport LocalCA when their dendrites use the same update policy as excitatory dendrites, $93.8 \pm 0.2\%$ when those inhibitory-cell updates are frozen, and $95.1 \pm 0.3\%$ under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder on the same one-layer architecture: rank-1 broadcast reaches $85.2 \pm 0.7\%$, while exact effective path transport reaches $95.2 \pm 0.0\%$. Thus explicit inhibitory cells can generate useful path-gain structure once the architecture isolates the dendritic-path question.

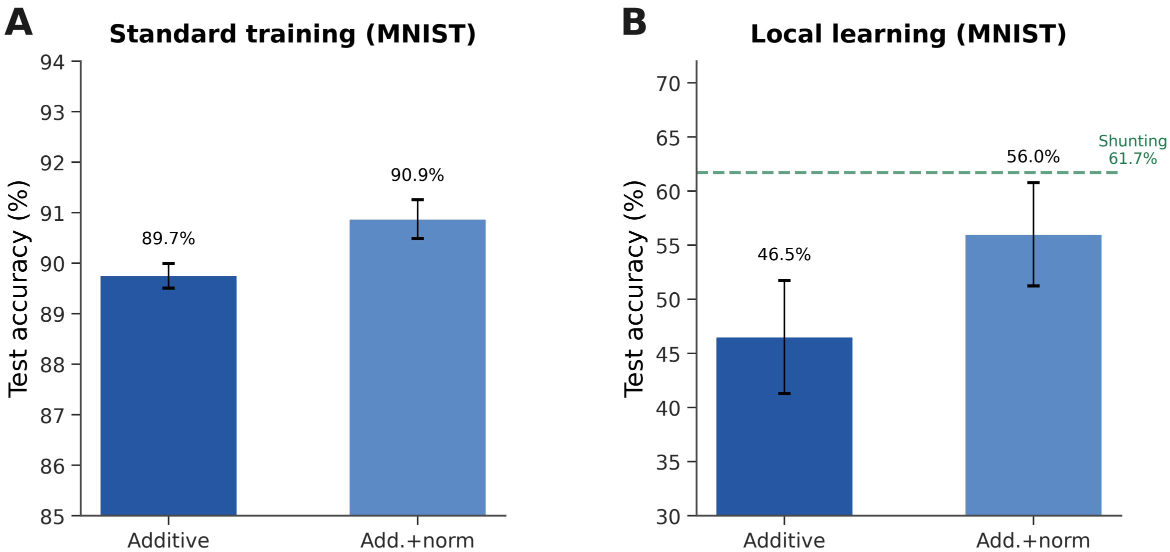

Figure S7: **Additive + normalization control (MNIST stress-regime diagnostic). (A)** Standard backprop: normalization provides a small boost (89.7% → 90.9%). **(B)** Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). These lower local accuracies come from a separate stress-regime diagnostic, not from the capacity-calibrated competence setting in Table S2.

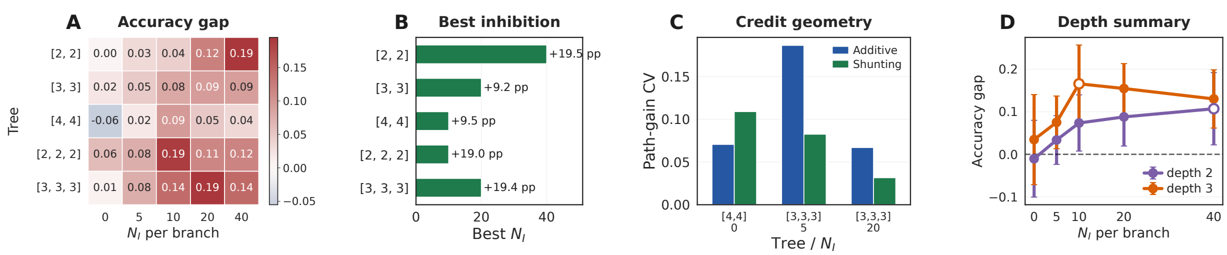

Figure S8: **Morphology shifts the useful inhibitory regime. (A)** Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. **(B)** Best inhibitory count for each morphology. **(C)** Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. **(D)** Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

Post-training inhibition intervention and exact-error rank. We next tested whether learned inhibitory conductance carries task-specific structure rather than only changing global scale. We replayed trained shunting LocalCA checkpoints while replacing the branch-level inhibitory conductance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST inhibition at N_I=5 gives 90.4 ± 1.1% accuracy, while the four interventions drop to 19.8–26.4%. On noise resilience at N_I=10, learned inhibition gives 80.5 ± 2.4%, while the interventions drop to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance load. We also computed the SVD of exact compartment-error matrices across samples and compartments. At matched MNIST N_I=5, shunting has lower rank-1 residual than additive (0.65 ± 0.18 vs. 0.83 ± 0.07) and lower participation rank (3.2 ± 1.2 vs. 8.5 ± 3.3), supporting the path-gain compressibility interpretation.

Inhibitory input mode	Training / broadcast	Test accuracy
Direct inhibitory stream	LocalCA, per-soma rank-1	0.852 ± 0.007
Direct inhibitory stream	LocalCA, exact effective path transport	$\mathbf{0.952 \pm 0.000}$
Explicit I cells	LocalCA, path transport, train I-cell dendrites	0.942 ± 0.002
Explicit I cells	LocalCA, path transport, freeze I-cell dendrites	0.938 ± 0.002
Explicit I cells	standard backprop	0.951 ± 0.003

Table S4: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with $[3,3]$ morphology and nonnegative inputs. The direct-stream rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with $[3,3]$ morphology. Exact effective path transport closes the direct-stream rank-1 gap, and explicit inhibitory cells nearly match both direct-stream path transport and the explicit-I backprop ceiling.

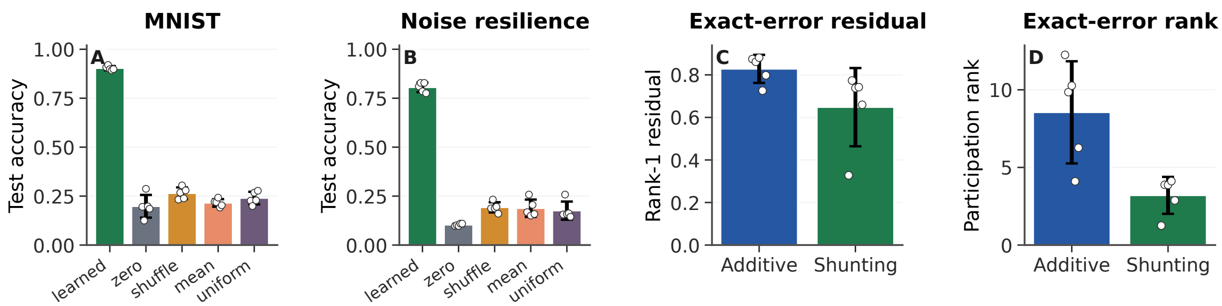

Figure S9: **Post-training inhibition intervention and exact-error compressibility. (A,B)** Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field carries sample-specific task structure. **(C,D)** SVD diagnostics of the exact compartment-error matrix on matched MNIST N_I=5 runs. Shunting has a smaller rank-1 residual and lower participation rank than additive, so the exact error field is more compressible. Error bars are s.d. across 5 seeds.

We also computed the direct log-gain covariance margin in Eq. (9) on the selected morphology diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly negative in the inhibited shunting selections (-2.5×10^{-3}, -5.0×10^{-4}, and -2.8×10^{-4} at N_I=5, 20, 40). We therefore do not use this covariance condition as a positive main result. Its role is to clarify when shunting should compress gains; the empirical claim rests on the measured comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

Cue-Routing Feedback-Rank Diagnostic

Cue routing tests the regime where one shared teaching signal is expected to fail: context determines which noisy cue is reliable, so different pathways should receive different credit. Figure S10 shows that the benchmark is learnable and that moving beyond rank-1 feedback helps, but the path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback method.

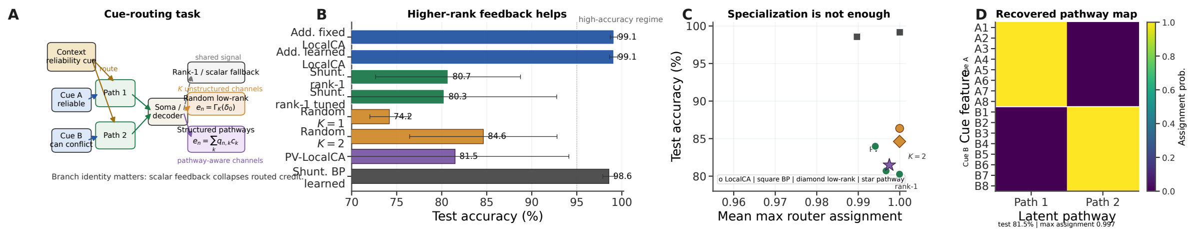

Figure S10: **Structured feedback when rank-1 broadcast fails. (A)** Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. **(B)** Five-seed comparison of rank-1, random low-rank, and pathway-structured feedback. Moving beyond rank-1 helps, while the best structured repair remains open. **(C)** Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. **(D)** Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

Setting	Broadcast	Test accuracy
Noise resilience, shunting, rank bridge		
per-soma	rank-1 per-soma	0.662 ± 0.016
path propagation	recursive attenuation proxy	0.762 ± 0.047
low-rank	$K{=}1$	0.453 ± 0.126
low-rank	$K{=}2$	0.630 ± 0.042
low-rank	$K{=}4$	0.764 ± 0.025
low-rank	$K{=}8$	0.795 ± 0.019
path transport	exact effective path transport	$\mathbf{0.847 \pm 0.018}$
Cue routing, learned shunting router, rank/structure sweep		
per-soma	tuned rank-1	0.803 ± 0.125
low-rank	$K{=}1$	0.742 ± 0.022
low-rank	$K{=}2$	$\mathbf{0.846 \pm 0.082}$
low-rank	$K{=}4$	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S5: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both recursive path propagation and higher-rank random low-rank broadcast close a substantial fraction of the gap between rank-1 per-soma and exact effective path transport, with low-rank continuing to improve up to $K{=}8$. On cue routing, the learned-router sweep shows that moving beyond rank-1 helps, but the best unstructured low-rank setting ($K{=}2$) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

Random Low-Rank Broadcast Channels

These controls test whether failures of rank-1 broadcast reflect insufficient feedback bandwidth rather than a failure of local eligibility itself.

B Implementation Details

Biological Plausibility Assumptions

The theorems and rules above rest on a small set of explicit assumptions. We state them as a checklist so that readers can see exactly what is assumed and where each assumption is used:

A1 (Steady-state voltage). We use the steady-state solution of the passive cable equation (Eq. (1)). Temporal dynamics are not modeled; all gradients, path gains, and fidelity measurements refer to the conductance-stage voltage at steady state.

A2 (Tree topology). Each dendritic cell is a rooted tree with the soma as the root. There are no recurrent connections within the dendrite, which makes the path from any compartment to the soma unique and is what allows the closed-form path gains α_n^{cond} and $\tilde{\alpha}_n$ in Theorem 1.

A3 (Nonnegative conductances). Synaptic and dendritic conductance parameters are pushed through a positive transform (`softplus`), so $g_j^{\text{syn}}, g_j^{\text{den}} \geq 0$ at every optimisation step. This is a biophysical constraint and is also what keeps the shunting denominator bounded away from zero.

A4 (Nonnegative presynaptic drive). The first synaptic drive x is nonnegative, implemented either by nonnegative inputs or an explicit ReLU transfer stage. Together with A3 this enforces the positive-input condition used for all main sweeps.

A5 (Pre-reactivation upward transfer). Theorem 1 is stated for the pre-reactivation compartment voltage. A post-voltage reactivation f (e.g. `param_tanh`) is handled by replacing α_n^{cond} with the effective gain $\tilde{\alpha}_n$ that includes $f'(V)$ factors along the path. The compartment error remains the sole non-local quantity; only the effective path gain acquires local activation-derivative factors.

A6 (Single perisomatic broadcast). Every synapse receives one non-local scalar or low-rank vector signal delivered from the soma; all other ingredients of the update rule (eligibility traces, driving force, input resistance, covariance modulators) are computable from quantities available at the synapse or compartment.

Each synapse therefore has access to: presynaptic activity x_j (local), compartment voltage V_n (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator r_n^{4F} requires online estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity is the broadcast error e_n, which requires a top-down or neuromodulatory signal from the soma to dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma broadcast mode.

Dendritic Structure

We use the morphology notation $[b_1, b_2, \ldots, b_D]$ for rooted dendritic trees, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3,3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward models, synaptic inputs terminate on dendritic branches rather than directly on the soma. The main setting uses direct input-driven inhibition: the transfer layer creates nonnegative excitatory and inhibitory streams from the same external input, and learned inhibitory conductance banks attach directly to excitatory dendritic branches. These models therefore have branch-level inhibitory conductance but no separate inhibitory neuron population. The explicit-I probe instead instantiates feedforward inhibitory cells,

Quantity	Symbol	Convention
Voltage	V	Conductance-stage shunting voltage in $[0,1]$ under nonnegative drive; additive controls may be signed
Conductances	$g^{\mathrm{syn}}, g^{\mathrm{den}}$	Nonnegative via $g = \mathrm{softplus}(\theta)$; local gradients are mapped to raw θ by the transform derivative
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S6: Units and normalization.

which transform excitatory activity before projecting inhibitory conductance back onto excitatory dendrites. In that setting the inhibitory cells are also dendritic modules and their synaptic conductances are trained by the same local rule family. The additive and shunting cores are architecture-matched at the tree level; they differ in the branch-level voltage integration rule rather than in whether a dendritic tree exists. In all reported dendritic runs, synaptic and dendritic conductance parameters are pushed through positive transforms, so conductances remain nonnegative; unconstrained decoder weights are separate readout parameters and are not interpreted as conductances.

Post-Voltage Reactivation

The conductance stage produces a pre-reactivation compartment voltage V_n, after which each branch layer may apply a monotone scalar nonlinearity. The main sweeps use a learnable bounded firing-rate transform, $(\tanh(m(V-b))+1)/2$, together with a nonnegative transfer stage so that the first synaptic drive remains nonnegative even after input corruption or structured task noise. Path-transport oracle runs with reactivation enabled use the effective transported error: parent-to-child transport includes the parent activation derivative, and the local update multiplies by the child derivative before applying conductance-space eligibility. Most main feedforward sweeps use a linear decoder, while the CIFAR extension uses a nonlinear decoder and therefore maps output error back to decoder-input/soma space through the decoder Jacobian before applying LocalCA. Unless explicitly overridden, additive and shunting cores can start from different reactivation initializations; activation-fairness ablations therefore match additive initialization to the shunting setting when comparing alternative reactivation choices directly. The appendix activation audit additionally fixes the bounded reactivation after initialization to isolate activation shape from activation plasticity.

Units and Parameterization

Decoder Update Modes

W_{dec} maps $V_L \to \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^\top \rangle_B$), **frozen** ($\Delta W = 0$).

Representative Manuscript Architectures

Hyperparameters

Table S8 collects the core training hyperparameters for each main experiment. Learning rates are applied through parameter groups (top-k, dendritic block-linear, reactivation, decoder) with the values given in the representative config files; where a single LR is reported, every group uses that value.

Effective Broadcast Dimensionality in Main Architectures

Compute Resources

All experiments were run on an institutional GPU cluster. Reproduction runs use one NVIDIA A100 or H100-class GPU (40–80GB memory) per seed and do not require distributed training. Table S10 gives approximate single-GPU runtime ranges for the reported experiment families; exploratory development runs outside the reported figures used additional cluster time and are not counted in the reproduction estimate.

Code, Data, and Asset Availability

MNIST, Fashion-MNIST, and CIFAR-10 are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. To preserve double-blind review, the submission provides code through an anonymized supplementary ZIP rather than a public repository URL. The ZIP contains the model, training, diagnostic, and figure-generation code together with representative configuration files and reproduction instructions for the main experimental families; code and configurations will be de-anonymized after review. The paper does not release a new dataset or a stand-alone pretrained model asset. Dataset access pages are the original MNIST site / UCI entry (`https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits`), the Fashion-MNIST repository (`https://github.com/zalandoresearch/fashion-mnist`), and the CIFAR page (`https://www.cs.toronto.edu/~kriz/cifar.html`).

Algorithm

Algorithm 1 Main 5F Rank-1 LocalCA Update

1: **Input:** Dendritic model, batch (x, y), learning rate η
2: Forward pass; loss L, output error δ^y
3: Somatic/core error $\delta_0 =$ decoder-input Jacobian product $J_{\mathrm{dec}}(h_{\mathrm{core}})^\top \delta^y$ (or $W_{\mathrm{dec}}^\top \delta^y$ for a linear decoder)
4: **for** each layer n (reverse) **do**
5:     Rank-1 broadcast $e_n = \mathrm{mean}(\delta_0)\mathbf{1}_n$
6:     Update EMA estimates for $r_n^{4\mathrm{F}}$ and ϕ_n from batch voltages
7:     $\Delta g_j^{\mathrm{syn}} \leftarrow \eta\, r_n^{4\mathrm{F}} \phi_n \langle x_j R_n^{\mathrm{tot}} (E_j - V_n) e_n \rangle_B$
8:     $\Delta g_j^{\mathrm{den}} \leftarrow \eta\, r_n^{4\mathrm{F}} \phi_n \langle R_n^{\mathrm{tot}} (V_j - V_n) e_n \rangle_B$
9: **end for**
10: Optional controls replace line 5 with neuron-wise, rank-K, path-structured, or transported-oracle broadcasts.
11: Clip gradients; optimizer step

C Theoretical Details

Random-Broadcast Alignment Intuition

Feedback-alignment arguments suggest that random or low-rank feedback can be useful when the induced local update remains positively correlated with the exact gradient [9]. In this model, that correlation depends on how local eligibility factors co-vary with the conductance-stage path gain in Eq. (4). We treat this as intuition only. The central theoretical object in the paper is still the exact pre-reactivation compartment error $\partial L / \partial V_n$; under identity upward transfer this reduces to $\alpha_n^{\mathrm{cond}} \delta_0$, while post-voltage activation uses $\tilde{\alpha}_n \delta_0$. The empirical question is how faithfully different broadcast fields approximate that quantity.

Morphology-Aware Extensions

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. (4).

Depth modulation. Per-branch scaling $\kappa_j = \kappa_{\text{base}}/(d_j + d_0)$, mirroring cable attenuation.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}}/(\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [20].

Pathway-vector feedback. For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

HSIC Auxiliary Objectives

Following [18], we use kernel-based HSIC objectives:

$$\mathscr{L}^{\text{self}} = B^{-2}\,\mathrm{tr}(\mathbf{K}_Z\mathbf{H}\mathbf{K}_Z\mathbf{H}), \qquad \mathscr{L}^{\text{target}} = -B^{-2}\,\mathrm{tr}(\mathbf{K}_Z\mathbf{H}\mathbf{K}_Y\mathbf{H}).$$

Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics ($r_n^{4\text{F}}$, ϕ_n) use Welford's algorithm [19].

D Task and Robustness Details

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. The MNIST-derived synthetic tasks use flattened 28×28 inputs; cue integration instead uses a small vector input with two output classes. Output dimensionality therefore varies by task and is stated in the task description when it differs from 10-way digit classification.

Context gating. Inputs are contextual figure-ground MNIST examples. The right half of the image carries the original MNIST digit, while the left half is replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. This task tests whether credit assignment can exploit a simple context-dependent spatial structure without leaving the nonnegative-drive regime of the conductance model. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are flattened MNIST images corrupted by structured channel noise: a fixed 50-dimensional latent Gaussian perturbation is projected into input space and added to each example, with $\sigma_{\text{task}} = 1.5$, then clipped to $[0,1]$. The projection is fixed across training, so the network must filter correlated interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

Cue integration. Two noisy cue streams (A and B) are presented simultaneously, each carrying partial information about one of 2 classes. A binary one-hot context signal indicates which cue is reliable on each trial (cue A or cue B), and each cue vector is clipped to $[0,1]$ after noise is added. The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in Sec. A to test structured pathway-vector broadcast while staying inside the nonnegative-input regime.

Depth Scaling and Noise Robustness

Depth scaling. Varying dendritic depth from 1–4 layers (branch factors [9] to $[3,3,3,3]$): shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth ($+8.5 \rightarrow +27.7$ pp). Backprop ceilings remain stable at about 90–92%. See Fig. S4A.

Noise robustness. Gaussian noise $\mathcal{N}(0,\sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about 62%) while additive drops from 46.5% to chance at $\sigma{=}1.0$, confirming that shunting credit signals carry useful learning information beyond the broadcast magnitude. See Fig. S4B.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	id.+ReLU	$[128, 128]$	$[3, 3]$	N_E=40, $N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 5 oracle	Main mechanism sweeps in Figs. 2 and 3; reactivation `param_tanh`.
MNIST / context-gating local competence	id.+ReLU	$[128]$	$[3, 3]$	N_E=40, N_I=20	5	Best local 5F per-soma runs in Table S2; reactivation `param_tanh`.
Fashion-MNIST competence	id.+ReLU	$[128]$	$[3, 3]$	N_E=40, N_I=20	5	Dedicated five-seed standard-vs-local competence sweep; reactivation `param_tanh`.
Cue-routing PV-LocalCA	router	$[64]$	$[2]$	N_E=10, N_I=4	5	Learned router with two pathway groups and rank-2 structured feedback; reactivation `param_tanh`.
CIFAR-10 compact direct-I-stream control ladder	id.+ReLU	$[20]$ E, no I cells	$[3, 3, 3, 3]$	N_E=25, N_I=25 direct I-to-E, with matched no-I controls	5 per condition	Harder-data extension with input-driven inhibitory conductance, decoder $[32, 16]$, decoder-aware soma mapping, and additive fairness controls.
Morphology$\times$inhibition appendix map	id.+ReLU	$[128, 128]$	$[2, 2]$, $[3, 3]$, $[4, 4]$, $[2, 2, 2]$, $[3, 3, 3]$	N_E=40, $N_I \in \{0, 5, 10, 20, 40\}$	3	Supportive 3-seed sweep; reactivation `param_tanh`.

Table S7: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks without direct somatic synapses, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

Setting	Opt.	LR	Batch	Epochs	W-dec.	Notes
MNIST / Fashion-MNIST / context gating (5F per-soma LocalCA)	Adam	10^{-3} / $5 \cdot 10^{-4}$ (block / react.)	256	100	0	fixed-epoch, no early stopping; `param_tanh` reactivation
MNIST / Fashion-MNIST / context gating (BP ceiling)	Adam	10^{-3} / $5 \cdot 10^{-4}$	256	100	0	matched to LocalCA setup
Gradient fidelity / path-gain / noise resilience	Adam	10^{-3}	256	50	0	5 seeds, hooks enabled for diagnostic capture
Morphology $\times$ inhibition regime map (supportive)	Adam	10^{-3}	256	50	0	3 seeds
Cue routing	Adam	10^{-3}	256	90	0	early stopping with patience 30
CIFAR-10 compact direct-I-stream depth-4 LocalCA	Adam	10^{-3} / 10^{-4} (block / react.)	256	400	0 / 0.01	grad-clip 5.0, early stop patience 50, nonlinear decoder
CIFAR-10 compact direct-I-stream depth-4 BP	Adam	10^{-3}	256	200	0.01	early stop patience 40, matched ceiling

Table S8: **Training hyperparameters by experiment.** All runs use Adam with default β_1=0.9, β_2=0.999, ε=10^{-8}. Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Representative configs are provided in the anonymized supplementary ZIP.

Setting	Hidden width(s)	Output dim	Effective default per-soma field
Gradient fidelity / path-gain / noise resilience	$128, 128$	10	Scalar / rank-1 at both dendritic layers
MNIST / Fashion-MNIST competence	128	10	Scalar / rank-1 at the dendritic hidden layer
Context gating competence	128	2	Scalar / rank-1 at the dendritic hidden layer
Cue routing	64	2	Scalar / rank-1 by default; higher-rank tests use explicit low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	decoder-input dim 20	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S9: **Effective dimensionality of the default neuron-wise broadcast in the main architectures.** The implementation uses the output error vector only when layer width matches decoder output dimension; otherwise it falls back to a shared scalar. The main feedforward settings in this paper therefore test a genuinely rank-1 broadcast regime by default, which is why the higher-rank low-rank, path-transport, and pathway-vector controls are important.

Experiment family	Runs	Time / run	Approx. GPU-h
Competence and verification	$\sim$25	5–20 min	4–9
Gradient, path-gain, inhibition, and oracle diagnostics	$\sim$80	10–30 min	15–40
Morphology, stress, rule, and feedback controls	$\sim$120	5–35 min	20–70
CIFAR-10 control ladder	70	0.8–2.5 h	60–175
Figure generation and CPU-side summaries	–	<5 h total	<5

Table S10: **Approximate compute for reported results.** Ranges reflect single-seed wall-clock variation across A100/H100-class GPUs, data-loading overhead, and early stopping.

Asset	Use in this paper	Access / license or terms note
MNIST	Digit classification and derived nonnegative synthetic tasks	Public benchmark from the original MNIST site / UCI entry; UCI lists DOI 10.24432/C53K8Q and asks users to follow the original acknowledgement policy; no raw-data redistribution.
Fashion-MNIST	Apparel classification stress test	Public Zalando Research benchmark; repository is MIT licensed; no raw-data redistribution beyond standard dataset loaders.
CIFAR-10	Flattened harder-data diagnostic	Public CIFAR dataset site asks users to cite Krizhevsky's technical report; no raw-data redistribution.
Synthetic tasks	Context gating, noise resilience, cue integration	Generated procedurally from public benchmarks or random seeds described in Appendix D; no new third-party asset is introduced.

Table S11: **Existing assets used in the manuscript.** We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the anonymous submission.